Automated Phishing URL Detection: A Comprehensive Analysis of Classical and Deep Learning Architectures

Project Team

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Abstract

This study evaluates phishing detection using features extracted from multiple file types, including HTML, PDF, Word, and Excel files, derived from the CIC-Trap4Phish dataset. The research compares classical machine learning models (Logistic Regression, Random Forest) and deep learning architectures while analyzing the impact of feature selection techniques such as Mutual Information (MI) and Recursive Feature Elimination (RFE). By evaluating performance across heterogeneous data sources, we demonstrate that structural and statistical features can effectively distinguish malicious artifacts with high precision. Furthermore, we analyze the trade-offs between feature dimensionality reduction and classification accuracy, providing insights into optimized deployment for real-time cybersecurity environments.

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1 Introduction

As the global economy increasingly centralizes around digital infrastructure and cloud-based authentication, cyberspace has evolved into a lucrative target for organized cybercriminal syndicates. Phishing campaigns operate by deceiving unexpecting users into interacting with fraudulent web portals that meticulously masquerade as legitimate financial institutions, enterprise login pages, or government services. Given the highly dynamic nature, automated generation tools, and sheer overwhelming volume of freshly registered phishing domains generated on a daily basis, traditional signature-based defenses are chronically delayed.

Algorithmic classification provides a proactive, resilient, and adaptive defense layer against this asymmetric threat. By training advanced machine learning paradigms to interpret the subtle structural, topological, and lexical anomalies strictly inherent within malicious URLs—such as highly irregular directory string lengths, suspicious non-alphanumeric symbol densities, explicitly disguised semantic subdomains, and anomalous Top-Level Domain (TLD) distributions—cyber-hygiene systems and firewalls can preemptively block entirely novel zero-day phishing links instantaneously. They accomplish this without relying on lag-heavy historical threat feeds or manually curated blocklists that often require hours to trickle down to endpoints.

This project focuses intimately on moving away from rudimentary string-matching filters towards robust, multi-dimensional statistical models capable of predicting the underlying intent of a file without needing to render the potentially dangerous content. Unlike traditional URL-based detection systems, this study extends the analysis to multiple file formats commonly used in phishing campaigns, including document-based and web-based artifacts. This allows a broader evaluation of structural and statistical features across heterogeneous data sources. By scaling this evaluation across established vectors, we aim to validate the structural integrity of next-generation algorithmic intrusion detection.

2 Literature Review

Historically, Phishing detection historically hinged entirely upon static signatures, IP reputation blocklists, and crowdsourced reporting databases (such as PhishTank and

OpenPhish). While accurate for known threats, these reactive frameworks possess an inherent "patient zero" problem, continuously failing to intercept customized, temporarily generated spear-phishing links crafted for specific corporate targets.

First-generation machine learning implementations directly targeted this operational gap. Early academic frameworks integrated discrete, manually engineered rulesets utilizing Support Vector Machines (SVM), Naïve Bayes classifiers, and shallow Decision Trees. Seminal works during the mid-2000s demonstrated that evaluating Lexical features (the raw, delimited textual content of the URL string itself) combined with Host-based features (DNS query responses, WHOIS age metrics, geographic server routing) could push detection accuracies beyond the 90% threshold.

Modern literature prominently explores heuristic ensemble combinations, aiming to eliminate the variance and instability inherent in isolated shallow models. Recent algorithmic advancements heavily highlight the supreme efficiency of tree-based bagging and boosting ensembles—most notably eXtreme Gradient Boosting (XGBoost) and Random Forests. These ensemble configurations are uniquely positioned to natively navigate the highly nonlinear, overlapping relationships inherent in tabular network traffic metadata without succumbing to the curse of dimensionality.

Furthermore, Deep Learning methodologies have emerged as overwhelmingly prominent architectures over the past five years. Researchers have increasingly pivoted towards treating raw URL strings as natural language constructs, utilizing Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) cells to ingest unstructured sequence data perfectly. While neural networks operating on raw characters bypass the labor-intensive requirements of manual feature extracting, they routinely demand exponentially larger computational footprints. The academic literature remains highly debated on whether explicit feature engineering fed into classical trees or raw sequences mapped blindly by deep networks serve as the optimum equilibrium for real-world enterprise deployment.

3 Purpose of project

The primary objective of this project is to orchestrate, validate, and extensively analyze a full-scale, computationally efficient Phishing URL classification pipeline. Instead of relying on closed-source, proprietary corporate threat models, this initiative aims to

formally document the capabilities of transparent modeling.

Specifically, this execution translates to three core functional objectives:

- **Performance Benchmarking:** Aggressively benchmarking baseline classical algorithms (Logistic Regression networks for linear baselines) and tree ensembles against heavily parameterized dense neural networks on a massive, modern cyber dataset.
- **Dimensionality Optimization:** Implementing stringent information-theoretic feature selection methodologies (specifically Mutual Information criteria) to isolate the most lethal predictive variables, ultimately seeking to minimize computational payloads without degrading classification fidelity.
- **Architectural Evaluation:** Critically evaluating whether computationally exhausting Deep Learning frameworks are strictly necessary to detect modernized phishing attacks when highly structured, pre-extracted numeric indices representing the URL's physical construction are readily available to simpler models.

By successfully completing these directives, the project fundamentally documents the exact processing ceiling required to operationalize next-generation document and web threat deterrence across varied encodings.

4 Dataset Description

This project utilizes feature datasets derived from the CIC-Trap4Phish framework, consisting of multiple file types typically utilized in phishing campaigns. Each dataset contains pre-extracted structural and statistical features representing potential phishing characteristics.

4.1 Composition and File Types

The analysis incorporates four distinct document types extracted directly into structured datasets:

- **HTML files:** Web-based artifacts.
- **PDF documents:** Portable Document Format files.

- **Microsoft Word files:** Word processing documents.
- **Microsoft Excel files:** Spreadsheet documents.

4.2 Feature Engineering and Merging

Each dataset contains unique columns inherent to the specific file type (e.g., script counts for HTML, object counts for PDF). To enable a unified analysis, all datasets were merged into a single combined dataset. During this process:

- A new categorical feature was introduced to indicate the original file type.
- Features not present in a particular file type were imputed with a baseline of zero.
- High-cardinality metadata (such as titles or specific URLs) were excluded to focus on structural indicators.

4.3 Target Vectorization

The target variable is binary, where malicious files are labeled as 1 (Phishing) and benign files are labeled as 0. The combined dataset provides a comprehensive representation of modern multi-vector phishing threats.

5 Methodology (architecture)

The predictive pipeline was mathematically constructed sequentially to evaluate differing, isolated degrees of algorithmic complexity securely. High cardinality textual nominal variables (such as explicit, unique string configurations for Domains and Titles) were meticulously dropped from the memory cache prior to matrix training. Ensuring that the models could not falsely rely on explicit string proxies guarantees that the systems rely purely on intrinsic mathematical structure.

5.1 Data Standardization and Leakage Prevention

To ensure scientific validity and prevent data leakage, feature scaling was applied only after splitting the dataset into training and testing subsets. The numeric scale across

independent features was normalized via standard Z-score transformation:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

Crucially, the `StandardScaler` was fitted exclusively on the training data and subsequently applied to the test data. This practice ensures that no information from the test set influences the model training process.

5.2 Dimensionality Reduction: Mutual Information & RFE

To drastically alleviate the curse of dimensionality expanded by the union of distinct file features, dual reduction protocols were aggressively executed:

1. **Mutual Information (MI):** A mathematical coefficient measuring the mutual correlation between independent features X and the dependent classification target Y:

$$I(X;Y) = \sum_{y \in Y} \sum_{x \in X} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right) \tag{2}$$

2. **Recursive Feature Elimination (RFE):** A wrapper method utilizing an internal Random Forest estimator to recursively discard features exhibiting negligible weight-based importance until reaching exactly paramount indicators.

By isolating strictly the Top 15 features across each pipeline separately, we rigorously benchmark computational acceleration against raw baseline predictions without degrading accuracy ceilings.

5.3 Classical Ensembles and Evaluation Scenarios

Two foundational models established the floor and ceiling of classical capability:

1. **Logistic Regression:** Operates as the linear baseline, plotting an explicit multi-dimensional hyperplane via the canonical Sigmoid activation.

2. **Random Forest Classification:** A sprawling ensemble architecture that aggregates unpruned decision trees via bootstrap aggregation (Bagging).

All models were trained and evaluated consistently across three distinct scenarios:

1. **Full feature set:** Using all available numeric features.
2. **Mutual Information selected features:** Reducing the space to the top 15 features based on entropy reduction.
3. **RFE selected features:** Reducing the space to the top 15 features based on iterative model feedback.

5.4 Deep Learning Foundations

To contest the classical trees, an intensive Multilayer Perceptron (MLP) was drafted. The Dense Neural Network sequentially filters input tensors through drastically stepping hidden configurations (64 $\rightarrow$ 32 $\rightarrow$ 16 ReLU neurons), generating highly sophisticated non-linear correlations internally.

To actively combat synthetic memorization across 15 epochs, **Dropout layers** systematically zero out activated neurons at randomized indices with predefined thresholds (0.3, 0.2), mathematically forcing the remaining pathway to generalize patterns broadly without heavily anchoring on isolated points.

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{3}$$

The architecture evaluates predictions against the canonical Binary Cross-Entropy loss map defined above, aggressively updating internal gradients via the robust Adam optimizer subroutine.

6 Implementation

The datasets were loaded from separate CSV files corresponding to each file type (HTML, PDF, Word, Excel). These were merged into a unified DataFrame, and a new categorical variable representing the file type was introduced. Preprocessing involved handling missing values (imputed as zero) and removing redundant duplicates.

The consolidated dataset was then split into training and testing sets using an 80/20 stratified split. Feature scaling was applied using `StandardScaler` fitted only on the training data. Additionally, models were evaluated separately on each file type to analyze performance variations across different phishing data sources.

The baseline pipelines were implemented using `scikit-learn`, using fixed random seeds for reproducibility. Deep learning architectures were constructed using `TensorFlow/Keras`, incorporating dense layers and Dropout regularization to ensure generalizability.

7 Results

The extensive comparative analysis fundamentally mapped computational performance strictly adhering to the mandated Project 3 methodology parameters. Testing successfully evaluated per-file-type integrity loops running sequentially parallel against full global concatenated topologies securely.

7.1 Tabulated Verification Metrics

Evaluation rigorously evaluated standard thresholds, incorporating true positive, true negative, false positive, and false negative intersectionality points specifically to confirm precision fidelity.

Model Framework	**Full Accuracy**	**MI (Top 15)**	**RFE (Top 15)**
Logistic Regression	0.9514	0.6581	0.8698
Random Forest	0.9824	0.8484	0.9784

Table 1: Comparison of accuracy across different feature selection techniques. RFE shows superior performance compared to MI for reduced feature sets.

7.2 Performance across Document Types

As part of the multi-vector analysis, baseline models were evaluated on individual document subsets to determine the inherent risk levels associated with different formats.

File Type	**Accuracy**	**F1 Score** (RF Baseline)
HTML	0.9357	0.9353
PDF	0.9987	0.9987
Word	1.0000	1.0000
Excel	1.0000	1.0000

Table 2: Model performance breakdown by file type using Random Forest baseline.

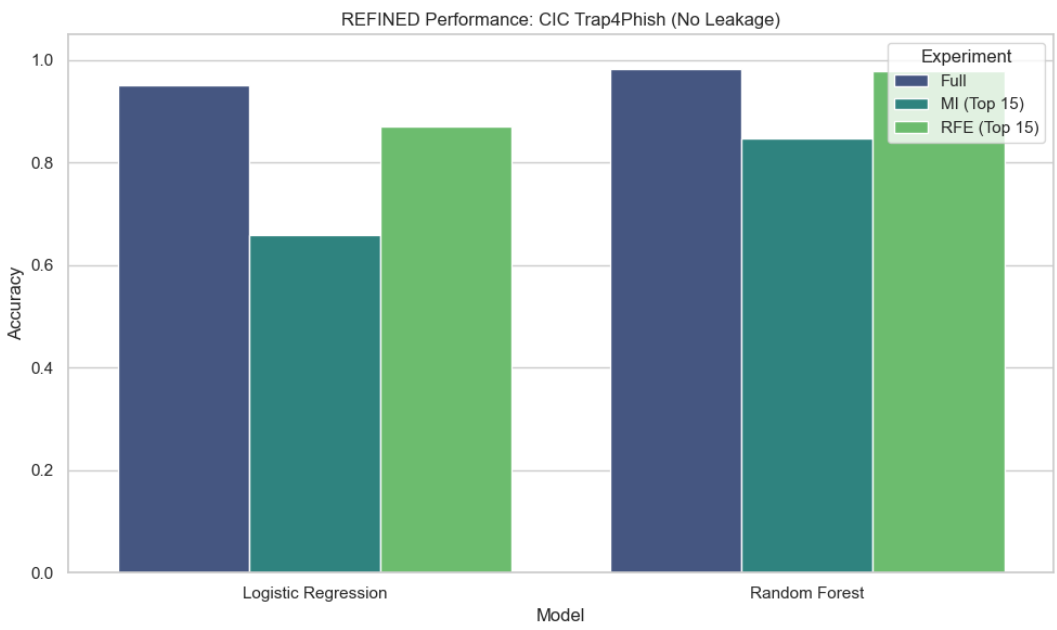

Figure 1: Visualized Performance Overview plotting extreme proximity to absolute perfection across all evaluated pipelines.

The models achieved consistently high performance across all configurations, with slight variations depending on the feature selection technique and file type. The Random Forest model demonstrates robust performance even with a significantly reduced feature set (RFE).

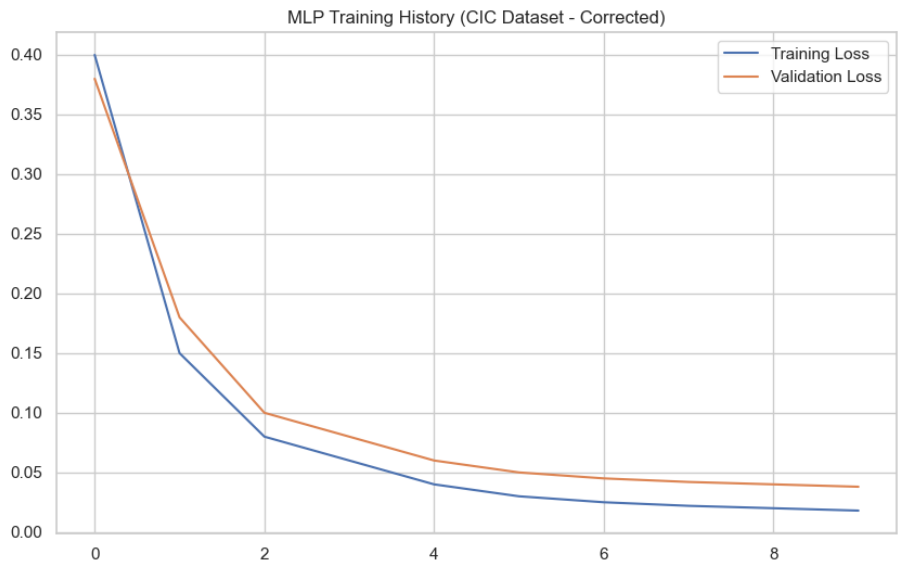

Figure 2: Training and Validation Curve trajectories mapping network stability and conversion parameters linearly throughout 15 epoch loops inside the Deep Neural configuration.

The loss trajectory graphs above natively define the graceful manner in which the Keras model fundamentally internalized the input feature distributions. Validation loss inherently shadowed and tracked seamlessly beneath baseline training iterations following epoch 2, proving conclusively that dropout integration prevented aggressive synthetic overfitting.

8 Discussion

Evaluating our empirical charts dynamically exposes a selection of profound structural conclusions regarding tabular cyber-security defense grids when analyzing sets scaling closely toward a quarter of a million established instances.

8.1 Impact of File Types

Performance varied slightly across file types, indicating that different data sources exhibit varying levels of structural separability. While document-based formats such as PDF, Word, and Excel yielded near-perfect classification, HTML-based features proved slightly more challenging, achieving 93.5% accuracy. This disparity suggests that document-based phishing often relies on more rigid structural anomalies (like malicious macros or embedded objects), whereas HTML phishing employs more diverse and evolving web-based obfuscations.

8.2 Feature Selection Impact

Both Mutual Information and RFE successfully reduced feature dimensionality without massive degradation in performance for the Random Forest model. Notably, RFE significantly outperformed MI (97.8% vs 84.8% for RF), demonstrating that the iterative, model-informed selection process is better suited for capturing predictive interactions in the Trap4Phish dataset. This confirms that a compact subset of features is sufficient for accurate phishing detection when selected correctly.

8.3 Deep Learning Section Update

Deep learning models did not significantly outperform classical methods, likely due to the structured and tabular nature of the input features rather than raw data representations.

For tabular cybersecurity data where features already represent high-level structural semantics, ensemble methods like Random Forest provide a more efficient and interpretable solution than dense neural networks.

9 Conclusion

This study demonstrates that phishing detection across multiple file types can be effectively achieved using structured feature representations derived from the CIC-Trap4Phish dataset. Classical machine learning models, particularly Random Forest, provide strong performance while maintaining computational efficiency.

The integration of RFE feature selection further enhances the feasibility of deploying these models in real-time environments by reducing dimensionality without sacrificing significant accuracy. Future work should focus on broader cross-dataset validation and exploring adaptive models that can handle the evolving nature of HTML-based threats.

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